Stochastic Processes 2021/22 Semester 1

Tutorial 2

1. Let $\{S_n\}_{n\geq 0}$ be a simple random walk with $p = 2/3$ (so that $q = 1/3$) and $S_0 = 0$.

   (a) Calculate $\mathbf{P}(S_{10} \geq 0)$.

   (b) Find an approximation for $\mathbf{P}(S_{100} \geq 30)$.

   (c) Find $\mathbf{P}$(there exists $n > 0$ such that $S_n \leq -2$).

2. An insurance company has initial capital u. Each year it receives premium income of 1 and pays claims. The total claim paid in each year i is Y_i where $\{Y_i\}_{i\geq 1}$ is a sequence of independent identically distributed random variables, each with the following distribution:

$$\mathbf{P}(Y_i = 0) = \frac{1}{2}; \quad \mathbf{P}(Y_i = 1) = \frac{1}{4}; \quad \mathbf{P}(Y_i = 2) = \frac{1}{4}.$$

   Calculate the value of u given that the probability that the company ever runs out of capital is 0.0078.

3. An insurance company has initial capital 5. Each year it receives premium income of 2 and pays claims. The total claim paid in each year i is Y_i where $\{Y_i\}_{i\geq 1}$ is a sequence of independent identically distributed random variables, each with the following distribution:

$$\mathbf{P}(Y_i = 0) = \frac{1}{20}; \quad \mathbf{P}(Y_i = 1) = \frac{7}{20}; \quad \mathbf{P}(Y_i = 2) = \frac{5}{20}; \quad \mathbf{P}(Y_i = 3) = \frac{7}{20}.$$

   (a) Explain why the probability that the company ever runs out of capital is **at most** $(7/8)^5$.

   (b) Let π denote the probability that the company runs out of capital within 100 years. In 100 simulations of this process, the number of times the company was ruined in 100 years was 29.

      i. Estimate π.

      ii. Calculate an approximate standard deviation of your estimate.

      iii. Estimate the number of simulations you would need to carry out for the standard deviation of your estimate to be ≤ 0.01.

4. Let $\{S_n\}_{n\geq 0}$ be a symmetric simple random walk (i.e. $p = 1/2$) with $S_0 = a \geq 0$. Let integer $b \geq a$ and let $T = \min_{n\geq 0}\{n : S_n = 0 \text{ or } S_n = b\}$ be the first time that the random walk hits either the state 0 or the state b. Define $e_{b,a} = \mathbf{E}(T)$.

   (a) Obtain a recurrence relation relating $e_{b,a}$ to $e_{b,(a+1)}$ and $e_{b,(a-1)}$.
   (**Hint:** Do not forget the term 1 to allow for the progression of time).

   (b) Verify that

$$e_{b,a} = a(b - a)$$

   satisfies the recurrence relation you have just attained and also the boundary conditions $e_{b,0} = e_{b,b} = 0$.

   (c) Deduce that, for $a > 0$, the expectation of the time to ruin is infinite (although ruin is certain in this case).